

Creating the Alien Class

Now we'll write the Alien class and save it as *alien.py*:

```
alien.py import pygame
        from pygame.sprite import Sprite

        class Alien(Sprite):
            """A class to represent a single alien in the fleet."""

            def __init__(self, ai_game):
                """Initialize the alien and set its starting position."""
                super().__init__()
                self.screen = ai_game.screen

                # Load the alien image and set its rect attribute.
                self.image = pygame.image.load('images/alien.bmp')
                self.rect = self.image.get_rect()

                # Start each new alien near the top left of the screen.
                ❶ self.rect.x = self.rect.width
                  self.rect.y = self.rect.height

                # Store the alien's exact horizontal position.
                ❷ self.x = float(self.rect.x)
```

Most of this class is like the Ship class, except for the alien's placement on the screen. We initially place each alien near the top-left corner of the screen; we add a space to the left of it that's equal to the alien's width and a space above it equal to its height ❶, so it's easy to see. We're mainly concerned with the aliens' horizontal speed, so we'll track the horizontal position of each alien precisely ❷.

This Alien class doesn't need a method for drawing it to the screen; instead, we'll use a Pygame group method that automatically draws all the elements of a group to the screen.

Creating an Instance of the Alien

We want to create an instance of Alien so we can see the first alien on the screen. Because it's part of our setup work, we'll add the code for this instance at the end of the `__init__()` method in *AlienInvasion*. Eventually, we'll create an entire fleet of aliens, which will be quite a bit of work, so we'll make a new helper method called `_create_fleet()`.

The order of methods in a class doesn't matter, as long as there's some consistency to how they're placed. I'll place `_create_fleet()` just before the `_update_screen()` method, but anywhere in *AlienInvasion* will work. First, we'll import the Alien class.

Here are the updated import statements for *alien_invasion.py*:

```
alien_invasion.py --snip--
                  from bullet import Bullet
                  from alien import Alien
```
